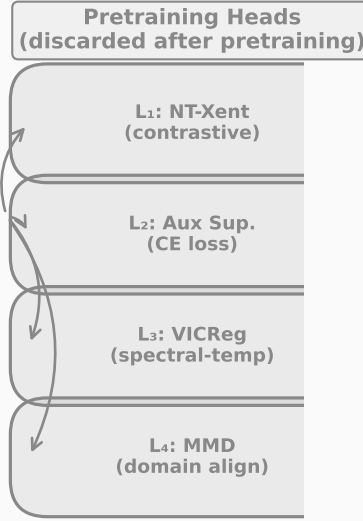
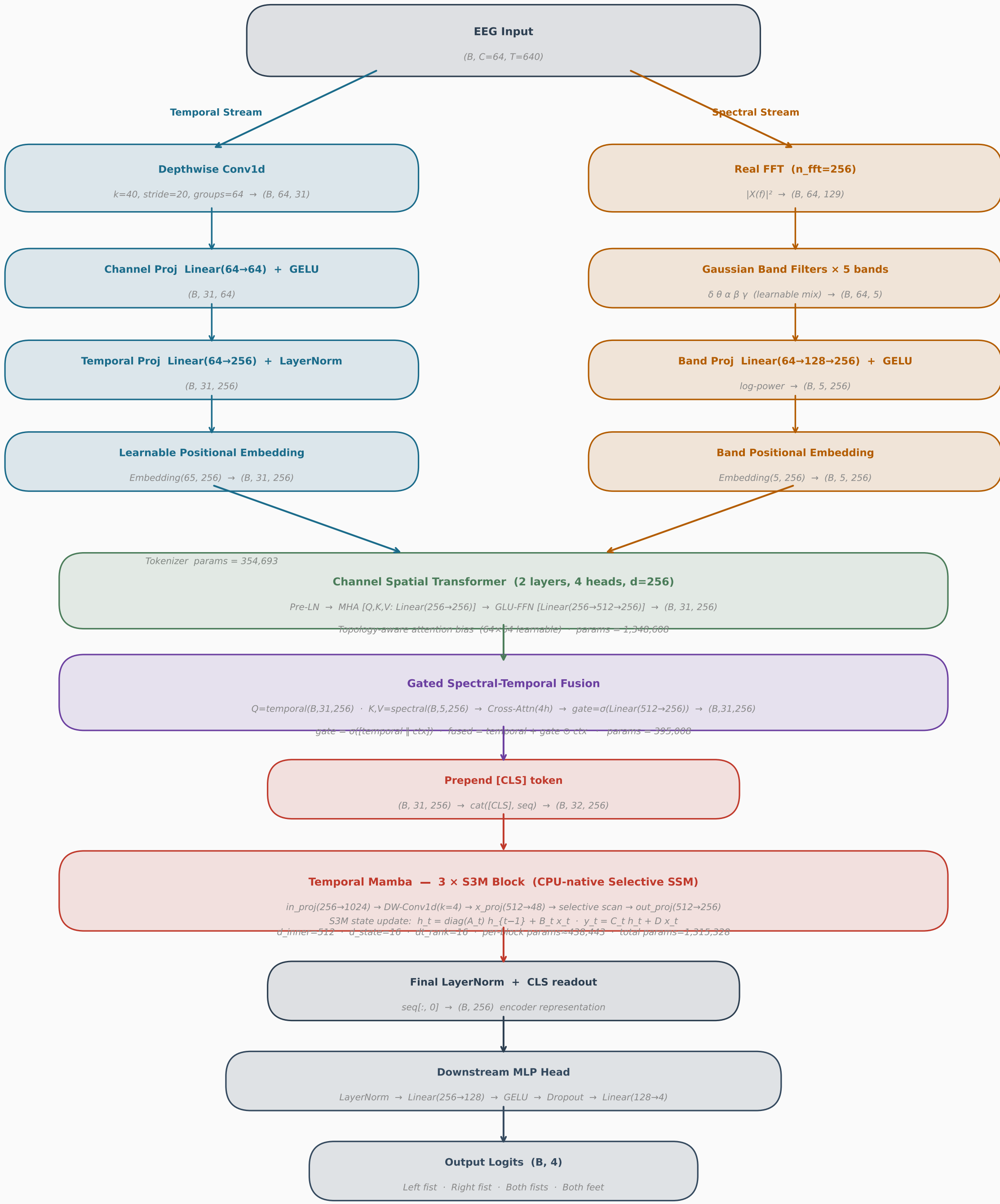


STELLA: Spectro-Temporal EEG Learning with Latent Alignment

Architecture Overview — PhysioNet EEGMMIDB (64ch, 160 Hz, 4-class MI)



Model Summary

Module	Output Shape	Params
Temporal Patch Embed	(B, 31, 256)	88,512
Spectral Band Encoder	(B, 5, 256)	47,941
Channel Spatial Transformer	(B, 31, 256)	1,348,608
Gated Spectral Fusion	(B, 31, 256)	395,008
S3M Temporal Mamba ×3	(B, 32, 256)	1,315,328
CLS + Final LayerNorm	(B, 256)	256 (CLS param)
Encoder Total	(B, 256)	3,414,405

Temporal Stream Spectral Stream Spatial Transformer Gated Fusion S3M / Mamba Classification Head

All operations are CPU-compatible · No CUDA dependencies · S3M: $O(L \cdot d \cdot N)$ sequential scan · Encoder: 3.41M params total